

REV2021

SEMESTER 2

PRINTING TECH

Office Automation Workshop

Course Code: 2109 | Engineering Science

[Official Syllabus PDF Source](#)

Course Overview

Program	Diploma in Printing Technology
Course Code	2109
Course Title	Office Automation Workshop
Credits	No Credit
Periods/Week	3 (L:0, T:0, P:3)
Total Periods	45

Course Objectives

- To provide knowledge of IT tools for professional word documents, spreadsheets, and presentations.
- To familiarize students with document preparation using office automation tools.

Course Outcomes (COs)

CO No.	Description	Hours	Cognitive Level
CO1	Perform word processing and documentation.	16	Applying
CO2	Perform spreadsheet preparation and accounting operations.	15	Applying

CO No.	Description	Hours	Cognitive Level
CO3	Develop power point presentation skills.	10	Applying

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	-	-	2	-	-	-
CO2	2	-	-	2	-	-	-
CO3	2	-	-	2	-	-	-

Module 1: Word Processing (CO1)

Duration: 16 Hours

M1.01: MS Word Orientation & Basics

Microsoft Word is a graphical word processing program that users can type with. Its purpose is to allow users to type and save documents.

- **Accessing:** Start > All Programs > Microsoft Office > MS Word.
- **Toolbars:** Quick Access Toolbar, Ribbon (Home, Insert, Page Layout, etc.).
- **Format Painter:** Used to copy formatting from one item to another. Select text with formatting -> Click Format Painter -> Select target text.
- **Rulers:** Horizontal and Vertical rulers help in aligning text, setting tabs, and margins.

- **Newspaper Columns:** Page Layout > Columns. Splits text into vertical blocks.
- **Table of Contents (TOC):** Automatically generated based on Heading Styles.
- **WordArt:** Decorative text that can be added to a document.

M1.05: Forms & Mail Merge

Mail Merge: A feature that allows you to create personalized letters or labels for a large number of recipients using a single template and a data source (like an Excel list).

Malayalam support: Mail Merge ഓട്ടോമാറ്റിക് ആയി അളവുകോലുകൾ ഉണ്ടാക്കാനും അതിൽ മാറ്റം വരുത്താനും സാധിക്കും. അതിനായി ഡാറ്റാ സ്രോതസ്സ് ഉപയോഗിക്കാം.

Module 2: Spreadsheets (CO2)

Duration: 15 Hours

M2.01 & M2.02: Excel Orientation & Scheduling

Excel is a spreadsheet program used to record and analyze numerical and statistical data.

- **Cell:** The intersection of a row and a column.
- **Auto Fill:** A feature that allows you to quickly fill a range of cells with a series of data (e.g., days of the week).

- **Summation:** Using the AutoSum (Σ) button to add values.

M2.03: Calculations & Functions

Function	Syntax	Description
AVERAGE	=AVERAGE(range)	Calculates the mean of values.
STDEV	=STDEV(range)	Calculates standard deviation.
VLOOKUP	=VLOOKUP(value, table, col_index, [range_lookup])	Searches for a value in the first column of a table.
COUNT	=COUNT(range)	Counts cells containing numbers.

Excel Logic Demo: Average Calculator

Value 1:

10

Value 2:

20

Calculate Average

Result: 15

M2.04 & M2.05: Performance Analysis & Pivot Tables

- **Freeze Panes:** Keeps specific rows or columns visible while scrolling.

- **Conditional Formatting:** Changes the appearance of cells based on specific conditions (e.g., highlighting marks > 40 in green).
- **Pivot Tables:** A tool used to summarize, analyze, explore, and present summary data.
- **Data Validation:** Restricts the type of data or the values that users enter into a cell.

Malayalam support: Pivot Table ഏകദേശം എളുപ്പത്തിൽ ഉപയോഗിക്കാൻ സഹായിക്കുന്നു. Conditional Formatting ഈ ഏകദേശം ഉപയോഗിക്കാൻ സഹായിക്കുന്നു. ഉപയോഗിക്കാൻ സഹായിക്കുന്നു.

Module 3: Presentations (C03)

Duration: 10 Hours

M3.01 & M3.02: PPT Orientation & Media

- **Slide Layouts:** Predefined arrangements for slide content (Title, Content, Two Content).
- **Auto Shapes:** Ready-made shapes like arrows, stars, and flowcharts.
- **Inserting Media:** Insert > Audio/Video. Enhances the presentation impact.

M3.03: Master Layouts & Animations

- **Slide Master:** The top slide that stores information about the theme and slide layouts. Editing the Slide Master affects all slides.

- **Slide Transition:** The visual effect that occurs when moving from one slide to the next.
- **Custom Animation:** Effects applied to individual elements on a slide (Entrance, Emphasis, Exit).
- **Auto Rehearsing:** Setting slide timings by practicing the presentation.

Malayalam support: Slide Master ഓരോ സ്ലൈഡ് ഓരോ സ്ലൈഡ് ഓരോ സ്ലൈഡ് ഓരോ സ്ലൈഡ് (ഓരോ സ്ലൈഡ് ഓരോ സ്ലൈഡ്) ഓരോ സ്ലൈഡ് ഓരോ സ്ലൈഡ്.

Formula Bank & Shortcuts

Excel Formulas

- =SUM(A1:A10) - Total
- =IF(A1>40, "Pass", "Fail") - Logical Test
- =MAX(B1:B20) - Highest Value
- =MIN(B1:B20) - Lowest Value

Common Shortcuts

- **Ctrl + N:** New Document
- **Ctrl + S:** Save
- **Ctrl + Z:** Undo
- **Ctrl + Y:** Redo
- **F5:** Start Presentation (PPT)

Visual Library

	Col A	Col B	Col C
Row 2		Active Cell	

Basic Spreadsheet Grid Anatomy

Module-wise Practice Questions

Module 1: Word

What is Mail Merge? (3 Marks)

Mail Merge is a feature in MS Word used to create multiple documents (like letters, envelopes, or labels) at once. It combines a 'Main Document' (template) with a 'Data Source' (list of names/addresses) to produce personalized outputs for each entry in the data source.

Explain the use of 'Format Painter'. (2 Marks)

Format Painter is used to copy formatting (font, size, color, etc.) from one piece of text and apply it to another. It saves time when you want consistent styling across a document.

Module 2: Excel

What is VLOOKUP? (5 Marks)

VLOOKUP stands for 'Vertical Lookup'. It is a function that searches for a specific value in the first column of a table range and returns a value in the same row from a specified column.

Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Module 3: PPT

What is a Slide Master? (3 Marks)

Slide Master is the primary slide in a hierarchy of slides that stores information about the theme and slide layouts of a presentation. Changes made to the Slide Master (like adding a logo or changing font) are automatically applied to all slides in the presentation.

Practice Model Paper

Time: 2 Hours | Max Marks: 50

Part A: Word Processing (20 Marks)

1. Create a professional resume using a table for educational qualifications. (10)
2. Design a newsletter with 3 columns and a drop cap for the first letter. (10)

Part B: Spreadsheets (20 Marks)

1. Create a student mark list for 10 students. Calculate Total, Average, and Result (using IF function). (10)
2. Generate a Bar Chart based on the mark list data. (10)

Part C: Presentation (10 Marks)

1. Create a 5-slide presentation on "Printing Technology". Include transitions and a slide master logo. (10)

Quick Revision

Word

- Drop Cap for style
- Mail Merge for bulk mail
- TOC for navigation

Excel

- Relative vs Absolute Ref
- Pivot Tables for summary
- Data Validation for accuracy

PPT

- Slide Master for consistency
- Transitions vs Animations
- F5 for slideshow

Glossary

Cell Referencing

Identifying a cell by its column letter and row number (e.g., A1).

Gutter Margin

Extra space added to the side or top margin of a document for binding.

Workbook

An Excel file that contains one or more worksheets.

Official Source Declaration

This lesson is developed based on the official **REV2021** syllabus for Course Code **2109** (Office Automation Workshop) provided by **SITTTR Kerala**.

Source: www.sitttrkerala.ac.in

മലയാളം പഠനസഹായം / Malayalam learning support

ഈ പാഠ്യപുസ്തകം സിറ്റിട്ര കേരളം നൽകിയ അനുമതിയിൽ തയ്യാറാക്കിയതാണ്.

- **MS Word:** 文字處理、排版、表格、圖表、郵件合併、宏、模板、安全、雲端協作。
- **MS Excel:** 試算表、數據分析、圖表、公式、數據透視表、宏、模板、安全、雲端協作。
- **MS PPT:** 簡報製作、排版、圖表、動畫、母片、模板、安全、雲端協作。

以上這些都是 Microsoft Office 軟體的實用知識 (Practical Knowledge) 範疇。